

Preliminary Mechanical Design of a Simple Robot Dog

1. Body Shape and Frame

The robot is designed as a simple four-legged robot dog. The body is rectangular and has four legs with a simple head at the front.

Dimensions:

- Length: 35 cm
- Width: 20 cm
- Height: 25 cm
- Weight: 3 kg

Material Used:

The frame is made of aluminum because it is lightweight and strong, which helps reduce the robot's weight and improve its stability.

2. Leg Design

The robot has four identical legs.

Each leg consists of:

- Hip Joint
- Knee Joint

This simple design allows the robot to stand and walk easily.

3. Number of Joints and Degrees of Freedom

Each leg moves in one basic plane and contains:

- Hip Joint = 1 DoF
- Knee Joint = 1 DoF

Total Degrees of Freedom:

$$4 \text{ legs} \times 2 \text{ Degrees of Freedom} = 8 \text{ DoF}$$

4. Motor Selection

The robot uses Servo Motors because they are easy to control and provide accurate rotational movement.

5. Initial Torque Calculation

- Mass on one leg (m) = 0.75 kg
- Distance from the joint (r) = 0.12 m
- Gravity (g) = 9.81 m/s²

First: Calculate the Force

$$F = m \times g$$

$$F = 0.75 \times 9.81$$

$$F = 7.36 \text{ N}$$

Second: Calculate the Torque

$$\tau = F \times r$$

$$\tau = 7.36 \times 0.12$$

$$\tau = 0.88 \text{ N}\cdot\text{m}$$

6. Stability and Center of Gravity

To improve the robot's stability:

- The battery is placed at the bottom center of the robot's body.
- The heavy components are placed close to the center of the body.
- The center of gravity should remain inside the area formed by the four legs while standing and walking.

7. Walking Pattern

Walking Sequence:

1. Front Right
2. Back Left
3. Front Left
4. Back Right

In each step, only one leg moves while the other three legs remain on the ground to maintain balance.

8. Expected Mechanical Problems

Some possible problems that the robot may face are:

- Incorrect joint alignment, which increases friction.
- High friction in the joints.
- Poor weight distribution, which may cause the robot to tilt.
- Loss of balance while walking on uneven surfaces.

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